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Reprint 2004-24 May 2004

DRAG COEFFICIENT DETERMINATION OF BLUFF BODIES – ANALYSIS OF BLOCKAGE EFFECT

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COST Action C14, Impact of Wind and Storm on City life and Built Environment, May 5-7, 2004, Rhode Saint Genèse, Belgium

DRAG COEFFICIENT DETERMINATION OF BLUFF BODIES – ANALYSIS OF BLOCKAGE EFFECT

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ABSTRACT: This paper explains how to correctly measure the drag coefficient of a circular cylinder in wind tunnels with large blockage ratios and for the sub-critical to the super-critical flow regimes. When dealing with large blockage ratios, the drag has to be corrected for wall constraints. Different formulations for correcting blockage effect are compared for each flow regime based on drag measurements of circular cylinders performed in a wind tunnel for three different blockage ratios. None of the correction models known in the literature is valid for all the flow regimes. To optimize the correction and reduce the scatter of the results, different correction models should be combined depending on the flow regimes. In the sub-critical regime, the best results are obtained using Allen and Vincenti's formula or Maskell's theory with $\varepsilon = 0.96$. In the super-critical regime, one should prefer using Glauert's formula with $G = 0.6$ or the model of Modi and El-Sherbiny. The change in the formulations is appearing at the flow transition with a variation of the wake pattern when passing from sub-critical to super-critical flow regimes. This parameter being not considered in the known blockage corrections, these theories are not valid for all the flow regimes.

1 INTRODUCTION

The von Karman Institute is performing an aerodynamic investigation to better understand the wind effect on the drag of octagonal poles, such as the light poles located along the roads. These poles have an octagonal cross-section with rounded edges. When produced, the poles are galvanized to be protected against corrosion. After years, the protection is less efficient and corrosion is appearing. This corrosion is modifying the surface roughness of the poles. The objective of that research contract is to experimentally investigate the effect of the rounded edge radius and of the surface roughness on the drag coefficient of the poles. This study is proposed and financed by the Ministère wallon de l'Equipement et des Transports through the "convention de services 451A.00.01". The authors are greatful to the Ministère wallon de l'Equipement et des Transports for authorizing them to publish and present the results obtained within the research contract.

The measurement of drag coefficient requires the use of very large wind tunnels in order to limit the wall effects on the measured results. As explained in section 4, wall effects are present for both open and closed wind tunnels and referred to as blockage effect. In practice, one can neglect blockage effect of bluff bodies when the wind tunnel cross-section is at least 30 times larger than the bluff body cross-section (blockage ratio smaller than 3%).

The drag coefficient is measured for Reynolds numbers, based on the cylinder diameter, ranging from $8 \cdot 10^4$ to $1.3 \cdot 10^6$ since the transition around cylinders is located within that range, as explained in section 2. This allows covering the sub-critical, the critical and the super-critical flow regimes. To cover the Reynolds number range, cylinders of different diameters are used, up to a diameter of 400 mm. The measurements are performed in the large low speed wind tunnel L1-A of the von Karman Institute, as explained in section 3. The maximum

velocity is 50 m/s and the test section has a height of 2.36 m, corresponding to a maximum blockage ratio of 17%.

In such conditions, the blockage effect cannot be neglected and corrections should be applied [1-7]. The objective of the present paper is to compare existing formulations to correct blockage effect on the drag coefficient measurements of bluff bodies in wind tunnels [5,8-12]. The bluff bodies considered in this paper are smooth circular cylinders.

2 DRAG COEFFICIENT OF SMOOTH CIRCULAR CYLINDERS

The flow around a circular cylinder has been widely investigated, some articles dating from the beginning of the fluid dynamics studies. Different kinds of transitions can occur in the wake of a bluff body, depending on the Reynolds number of the flow [13-16]. The Reynolds number is the non-dimensional parameter used to characterize the ratio between the inertia and the viscous forces:

$$Re = \frac{Ud}{\upsilon} \tag{1}$$

where U is the free stream velocity, d is the cylinder diameter and ν is the fluid kinematic viscosity. So, the Reynolds number characterizes the flow regime and thus the drag induced by the wind on the cylinder. The drag coefficient and the base pressure coefficient are defined as:

$$C_D = \frac{D}{\frac{1}{2}\rho U^2 S} \qquad C_{pb} = \frac{p_b - p_s}{\frac{1}{2}\rho U^2} \tag{2}$$

where D is the drag (aerodynamic force parallel to the wind direction), S is the projected section perpendicular to the wind direction, $\frac{1}{2}\rho U^2$ is the dynamic pressure of the wind (ρ is the fluid density), p_b is the base pressure and p_s is the static pressure.

The evolution of the drag coefficient with the Reynolds number for a circular cylinder appears in many references [13-19] and is given in Figure 1.

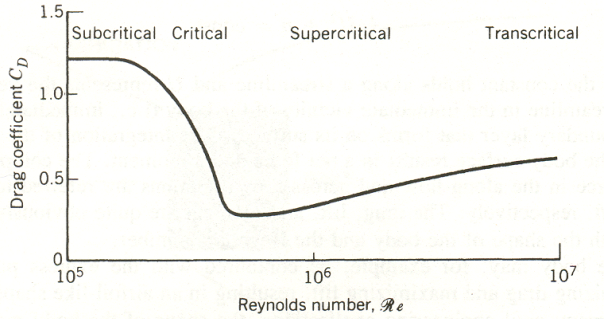


Figure 1: evolution of drag coefficient for circular cylinder [19,23].

Details about the flow field can be found in references [13-15,20-23]. Depending on the flow regime, the flow field downstream the cylinder is complex and presents different features. This complexity comes from the evolution of the position of the separation points along the cylinder circumference. Different authors have then defined ranges of Reynolds numbers within which the flow features could be well characterized [13,15]. The evolution from each regime to the next corresponds to a change in the position of the laminar to turbulent transition. This transition does not appear simultaneously in the whole wake, but as successive laminar to turbulent transitions initiated first in the far wake before moving back progressively to the cylinder surface. The different flow regimes are reported in Table 1 and schematized in Figure 2.

At very low Reynolds number ($Re < 1$), the flow remains viscous and attached all around the cylinder (Figure 2a). When Re increases, the flow separates from the cylinder surface to form a closed wake of recirculating flow (Figure 2b). The wake grows progressively in length as Re is increased from 5 up to 40. The flow pattern is steady and symmetrical. When Re exceeds a value of 40, steady flow becomes impossible and the steady laminar wake develops into an unsteady, but stable, laminar wake. Vortices are then periodically generated on alternate sides of the axis. This type of wake is termed as von Karman vortex street (Figure 2c).

Then the turbulence level increases in the wake, but the shear layers near the cylinder remain laminar. The separation points are still laminar and the transition appears in the wake far downstream from the cylinder (Figure 2d).

Figure 2	*Re*	Flow regime
(a)	0 – 5	Viscous flow without separation
(b)	5 – 40	Steady recirculating flow
(c)	40 – 400	Laminar separation - von Karman vortex street
(d)	$400 - 2{\cdot}10^5$	Sub-critical
(e)	$2{\cdot}10^5 - 5{\cdot}10^5$	Critical
	$5{\cdot}10^5 - 3{\cdot}10^6$	Super-critical
(f)	$3{\cdot}10^6$ –	Trans-critical

Table 1: the different flow regimes [13,15].

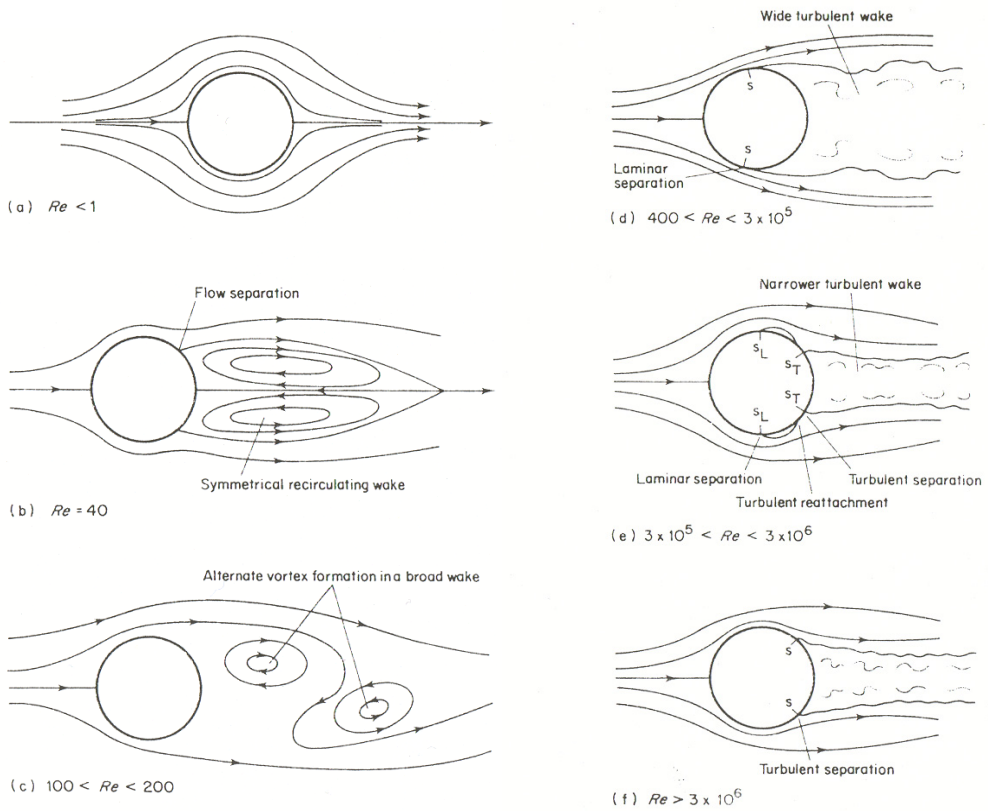


Figure 2: schematic of the different flow regimes [13].

When the Reynolds number increases, this transition goes closer to the cylinder surface up to the separation points. When *Re* reaches a value in the vicinity of $3{\cdot}10^5$ the laminar boundary layer undergoes transition to turbulence almost immediately after separation. The increased mixing re-energizes the separated flow causing it to reattach as a turbulent boundary layer (Figure 2e). A separation bubble is then formed. At this critical stage the separation point in the turbulent layer moves suddenly downstream, because of the better sticking property of the turbulent layer. This is accompanied by a sudden decrease in the total drag of the cylinder (Figure 1). For this reason, the value of *Re* at which this transition occurs is called the critical Reynolds number.

With further increase in Reynolds number, the turbulent separation points slowly move upstream along the boundary layer towards the stagnation point. This regime is called supercritical. The final change in the flow pattern occurs at $Re \cong 3{\cdot}10^6$ when the separation bubble disappears (Figure 2f).

The transitions correspond to distinct flow regimes that affect pressure distribution over the body, and then the resulting drag force. The critical regime appearing for Reynolds number around $Re = 3{\cdot}10^5$ is associated with a strong decrease of C_D. On both side of this critical regime, the drag coefficient is characterized by two constant values as shown in Figure 1 ($C_D = 1.2$ at the end of the sub-critical regime and $C_D = 0.3$ at the beginning of the supercritical regime). In practice, these two values correspond to the most critical ranges of Rey-

nolds numbers, in term of drag on the cylinder. Indeed, even if the drag coefficient at low Reynolds number ($Re < 100$) is high, the force acting on the cylinder is low. When Reynolds number is higher than 10^4, the effect of the velocity on the drag D becomes important, according to relation (2). The critical regime being associated to a strong decrease of drag coefficient, the only regimes for which the drag force is important are the sub-critical and the early super-critical regimes, characterized by constant C_D values. Later in the super-critical regime, both the Re and the C_D are still higher but the wind velocity is out of the common practical range (< 200 km/h).

The drag evolutions in the critical regime vary significantly from one author to another, as reported in the literature [19,22]. This could be explained by the high sensitivity of the boundary layer transition to external parameters, such as the cylinder roughness, the turbulence intensity of the upstream flow, the blockage ratio (ratio between the model and the wind tunnel cross-section areas) the aspect ratio of the model and the geometry of the end plates.

3 WIND TUNNEL EXPERIMENTS

The measurements have been performed in the free jet test section of the VKI low speed wind tunnel L-1A. The test section of 3 m diameter and 4,5 m length is adapted with a rectangular insert with solid side, top and bottom walls that is used for tests of 2-D models. The flow velocity can vary continuously from 2 to 50 m/s. The contraction ratio is 4 with a typical turbulence level of 0.3%. The insert is equipped with a transversal model support with strain gauge balances for the measurement of lift and drag (Figure 3). For the present work, tests have been made on three smooth cylindrical models with circular cross-sections of different diameters to vary the blockage ratio (Table 2). The three circular cylinders are in aerodynamic similitude.



Figure 3: the VKI low speed wind tunnel with rectangular insert equipped with strain gauge balances and one of the circular cylinder. (a) front view ; (b) back view.

Diameter *(d)*	Blockage ratio (*S/C*)	Re_{max}	Aspect ratio (*l/d*)
160 mm	6.8 %	$5.2 \cdot 10^5$	8.12
320 mm	13.6 %	$1.0 \cdot 10^6$	4.06
400 mm	17 %	$1.3 \cdot 10^6$	3.25

Table 2: characteristics of the smooth cylindrical models.

When testing in a wind tunnel, the model has a finite length and the model extremities and side walls could introduce secondary effects [24,25]. Slaouti and Gerrard [26] demonstrated that these effects have a strong influence on the near wake and then on the drag coefficient. To guarantee the bidimensionality of the flow around the cylinder, end plates are added perpendicularly to the cylinder axis. This is the reason for using the rectangular insert described just before. Chen and Doepker [7] showed that to guarantee the flow bidimensionality at high Reynolds numbers ($3.5 \cdot 10^5 < Re < 1.2 \cdot 10^6$), the cylinder aspect ratio should be higher than 3 times the model diameter. Achenbach [27] reported also that in the critical flow regime, there was an effect on the drag force for an aspect ratio lower than 3 and found no systematic influence between the aspect ratios of 3.33 and 6.66. In our experiments, the end plates are separated by 1.3 m, leading to aspect ratio provided in Table 2. The end plates need also to be very large. Their size influences strongly the length over which the flow can be considered as bidimensional. In our experiments, the ratio between the total length of the end plates (up-

stream and downstream the model) and the diameter of the cylinder ranges from 10 to 26 depending on the cylinder diameter. Finally, if the upstream or the downstream length of the end plates is too small, secondary effects could influence the upstream flow or the wake. In our experiments, the length of the end plates downstream the model is equal to 1.75 m, which is higher than the value of 4.25d suggested by Kubo et al. [28]. The upstream length of the end plates is equal to 2.5 m, which proved to be sufficient since a reference plane with uniform pressure distribution (static and dynamic) has been identified at 1.8 m upstream of the model, for the 3 cases tested.

The free stream reference flow velocity U was measured using a Pitot tube connected to a pressure transducer. The drag force D was measured using two strain gauges balances connected to both extremities of the cylinder. The instantaneous drag results are recorded on a PC, before being averaged. It was checked that the mean value is independent of the number of acquisition points.

Figure 4a shows the drag evolutions obtained for the three cylinders and the evolution obtained by Wieselberger [16]. One distinguishes the sub-critical, the critical and the early super-critical regimes. The drag force is increasing with the blockage ratio and should be corrected. The corresponding base pressure coefficients are given in Figure 4b. The difference observed between the curves of Figure 4 comes from the blockage effect.

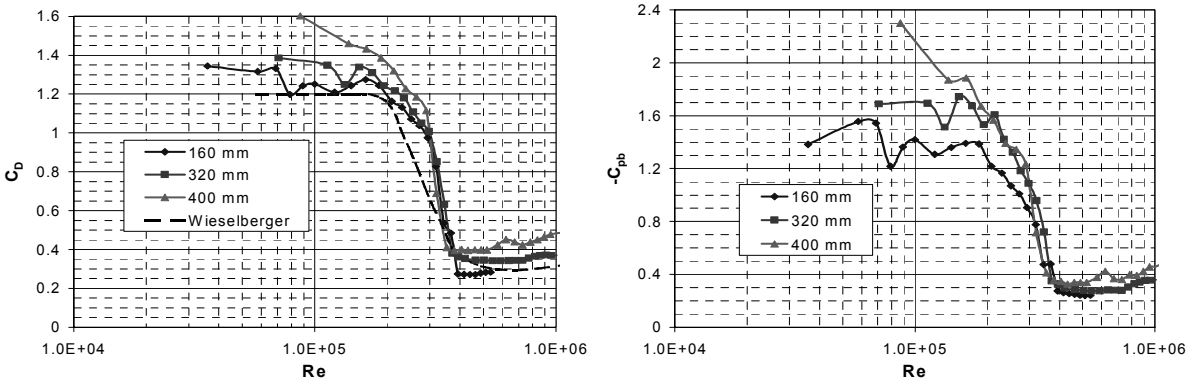


Figure 4: the uncorrected drag coefficient and uncorrected base pressure coefficient evolutions for circular cylinders in the VKI low speed wind tunnel.

4 BLOCKAGE EFFECT

In a closed test section, the lateral walls prevent the flow to expand when passing around the cylinder (Figure 5b). The flow around the cylinder is then accelerated compared to the value expected if the cylinder was put in an infinite space. When speaking about infinite space, one should consider that the cylinder is not surrounded by wall and that the velocity is injected over an infinite cross-section. In these infinite conditions, the flow over the cylinder will be referred to U_c (corrected velocity). The flow acceleration obtained for a closed test section generates an over-estimation of the drag coefficient ($C_D > C_{D,c}$).

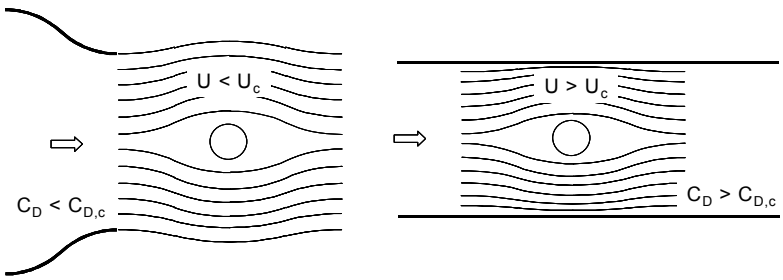


Figure 5: blockage effect in open and closed test section wind tunnels.

In an open test section wind tunnel, the model is placed in an "infinite" space. The flow can expand laterally (Figure 5a), but the velocity is only injected over a finite cross-section (the nozzle exit of the wind tunnel). This jet injection generates air entrainment from the surroundings. It results in an over-expansion of the flow around the cylinder ($U < U_c$) corresponding to an under-estimation of the drag coefficient.

Blockage corrections are necessary for both closed and open test section wind tunnels. Figure 6 shows a comparison of the closed and open test sections in the case of normal flat plates and rectangular blocks, both wall-mounted and centrally-mounted in the test section [1]. The models considered in this figure are three-dimensional bluff-bodies. As expected, the closed test section shows a drag increase with model area while the open test section shows a drag reduction. The blockage effect in the open test section is less than in the closed one. However it is more suitable to perform measurements in a closed test section since the correction formulations are better developed and the boundaries are more precisely defined in this case. The corrected drag coefficients obtained from a closed test section are certainly more reliable than those measured in an open test section without correction.

The present tests were performed by inserting the cylindrical models within a rectangular insert corresponding to a closed test section.

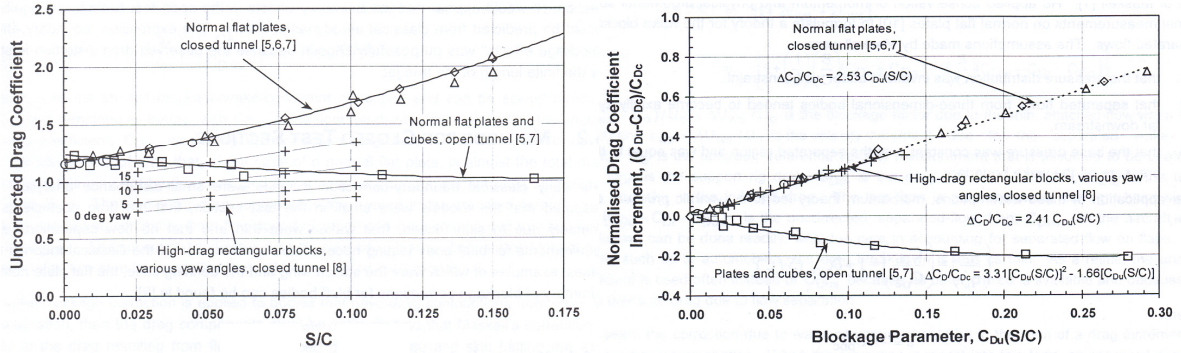


Figure 6: bluff-body blockage effects on drag in closed and open test sections [1].

4.1 Theories for Blockage Correction

Several authors have proposed formulations for blockage correction. A mathematical prediction of blockage effects on flow past cylinders in the closed test section involves a complete solution of the Navier-Stokes equations. However, most theoretical models are based on the inviscid potential flow theory or general momentum balance [29].

The three first theoretical corrections (Fage [8], Lock [9] and Glauert [10]) are derived for bodies of small drag such as aerofoils and not for high drag bodies like circular cylinders. They are based on the two-dimensional potential flow around Rankine's ovals (the circular cylinder appears as a special case) between straight and parallel walls. These correction formulations provide the ratio between the corrected and uncorrected drag coefficients. This ratio does not depend on the drag coefficient and on the base pressure coefficient but depends only on the blockage ratio and on an empirical factor. Within these three theories, the most advanced one is the theory of Glauert who suggested the next semi-empirical formula:

$$\frac{C_{D,c}}{C_D} = \left[\frac{1 - G(S/C)}{1 + 0.822(S/C)^2} \right]^2 \tag{3}$$

where $C_{D,c}$ is the corrected drag coefficient, S is the projected section perpendicular to the wind direction, C is the cross-section of the test section and G is an empirical factor. Glauert [10] found a value of $G = 0.3$ for circular cylinders. Modi et El-Sherbiny [11] applied Glauert's formula to their drag measurements and showed that changing the empirical factor to $G = 0.6$ gives a considerable improvement and reduces scatter of points. Figure 7a shows the corrected drag coefficient using Glauert's formula with $G = 0.6$. The scattering of the three curves is well reduced compared to raw data (Figure 4a). Relation (3) works well in the supercritical regime while it is over-correcting in the sub-critical regime.

Allen and Vincenti [12] derived a theoretical model for the wake blockage by considering an inviscid, compressible and two-dimensional flow in a closed wind tunnel. Two stations in the wind tunnel are considered, far upstream from the cylinder and far enough downstream so that the wake has spread to the walls and the velocity is again uniform across the tunnel. The difference in static pressure between the two stations is evaluated as a function of the measured cylinder drag. The tunnel walls are simulated by an infinite system of sources placed di-

rectly above and below the position of the cylinder at intervals equal to the breadth of the tunnel. The drag coefficient correction terms are all negative and tend to reduce the measured drag coefficient. The final expression is:

$$\frac{C_{D,c}}{C_D} = 1 - \frac{\pi^2}{4}\left(\frac{S}{C}\right)^2 - \frac{C_D}{2}\left(\frac{S}{C}\right) \tag{4}$$

Figure 7b shows the corrected drag coefficient using Allen and Vincenti's relation. In the super-critical regime, the scatter is higher than when using Glauert's formula (Figure 7a), while Allen and Vincenti's formula works much better in the sub-critical regime.

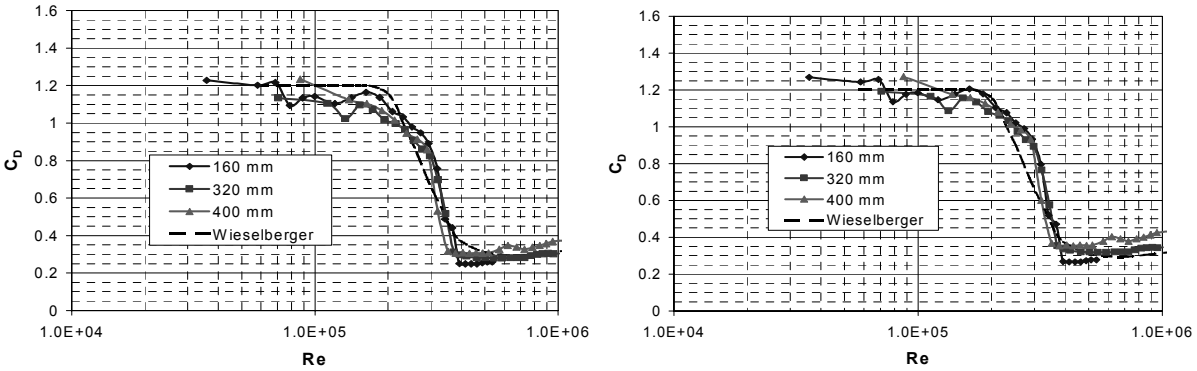


Figure 7: the drag coefficient corrected by Glauert's formula with G = 0.6 (a) and by Allen and Vincenti (b).

Maskell [5] has developed a semi-empirical model for axisymmetric wake and aerofoils. He also applied his theory to flat plates perpendicular to the flow, even for very high blockage ratios up to 20%. His theory is based on conservation of momentum between a cross-section upstream of the body where the flow is not perturbed and a downstream cross-section where the wake presents its maximum width. The assumptions made by Maskell are that:

- the pressure distribution is invariant under wall constraint,
- separated flows from three-dimensional bodies tend to become axially symmetric far downstream,
- the base pressure is constant over the separated region and is equal to the static pressure on the wake boundary,
- the constraining effect of the test section walls reduces the expansion of the wake and that this reduction is in proportion to the contraction of the external stream around the wake.

The final form of Maskell's theory is given by:

$$\frac{C_D}{C_{D,c}} = 1 + \varepsilon C_D \frac{S}{C} \tag{5}$$

where ε is the blockage constant. From best linear fit of experimental data, Maskell showed that $\varepsilon = 0.96$ for bidimensional bluff bodies with sharp edges and with a wake width of the order of the bluff body dimension. This coefficient is theoretically linked to the base pressure coefficient by:

$$\varepsilon = \frac{-1}{C_{pb}} \tag{6}$$

Using relation (5), it is possible to correct both the drag coefficient and the Reynolds number. But this assumes the use of the value 0.96 or the use of measured base pressure coefficients (relation 6) for the blockage constant. Both possibilities have been compared on the data of Figure 4 since the base pressure was measured. It was concluded that the use of Maskell's theory with blockage constant deduced from the base pressure coefficient (relation 6) is not providing better results compared to Allen and Vincenti's formula in the sub-critical regime and to Glauert's formula in the super-critical regime. Figure 8a shows the corrected drag coefficient using Maskell's theory with $\varepsilon = 0.96$. This value works well in the sub-critical re-

gime, the results being similar to those obtained with Allen and Vincenti's formula. However, scatter is observed in the super-critical regime proving that a single constant value for ε will never provide good results both in the sub-critical and super-critical regimes.

Modi and El-Sherbiny's blockage derivation [11] is based on the flow around a circle between parallel walls. The near-wake is simulated by two sources attached to the downstream side of the circle and by two sinks placed at the circle center. The source strength is determined by the location of the separation points. This model has been developed for two kinds of two-dimensional bodies: the flat plate and the circular cylinder. The experiments carried out to validate the model covered a wide range of blockage ratios (up to 35%) and Reynolds numbers ($10^4 < Re < 1.2 \cdot 10^5$). According to the authors, the theoretical model predicted the drag coefficient quite well, even for high values of blockage ratio, and the validity of the model does not seem to be limited to these values of Reynolds numbers, although this was not demonstrated because of a lack of experimental data in these conditions. The only limitation of the model is the concept of the wake without eddy street.

Figure 8b shows the corrected drag coefficient using Modi and El-Sherbiny's theory. The formula works well in the super-critical regime while it is over-correcting in the sub-critical regime. The conclusion is similar to the one derived from Glauert's formula with G = 0.6.

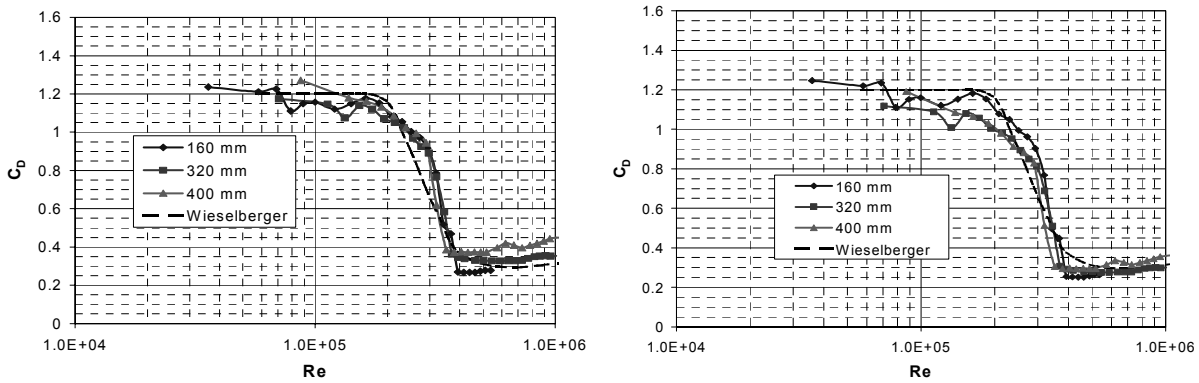


Figure 8: the drag coefficient corrected by Maskell with $\varepsilon = 0.96$ (a) and by Modi and El-Sherbiny (b).

4.2 Blockage effect at different flow regimes

The results from Figures 7 and 8 indicate that it is possible to correct drag coefficients when measured in a wind tunnel under high blockage ratio. However, to optimize the correction and reduce the scatter of the results, different correction models should be applied depending on the flow regimes. In the sub-critical regime, the best results are obtained using Allen and Vincenti's formula or Maskell's theory with $\varepsilon = 0.96$. In the super-critical regime, one should prefer using Glauert's formula with $G = 0.6$ or the model of Modi and El-Sherbiny.

It seems strange that none of the aforementioned model is valid for all the flow regimes. Indeed, the theories developed up to now are considering that the wall constraint depends only on the blockage ratio, the measured drag coefficient and an empirical factor, except Maskell that considers also the effect of the base pressure, when used with relation (6)[1]. So, all the theories aforementioned should provide an accurate correction for blockage effect whatever the flow regime since neither the Reynolds number nor the wake properties appear explicitly in the correction model. However, this is not the case for circular cylinders, for which we found a unique theory is not valid.

The theories, like the Maskell's theory with $\varepsilon = 0.96$, assume that the base pressure evolution is proportional to the drag coefficient evolution. So, as observed in Figure 4, if the drag coefficient is increased, the base pressure coefficient is increasing (in absolute value) so that

$$\frac{C_D}{1 - C_{pb}} = \text{constant} \tag{7}$$

[1] However, Maskell's theory, when based on the base pressure, has proved to fail for circular cylinders.

Figure 9 shows the evolution of the drag coefficient versus $(1 - C_{pb})$ for the three circular cylinders. The values of drag coefficients are located along two different slopes. The change in the slope is appearing for $C_D = 1$, which corresponds to the flow transition when passing from sub-critical to super-critical flow regimes. During that transition a variation of the wake pattern is occurring that is not considered as a parameter in the blockage corrections presented above. This might explain why the theories are not valid for all the flow regimes, since they are defined with only one empirical factor in place of one factor per slope as in Figure 9.

There is no reason indicating that the theories cannot be applied to the different flow regimes but the empirical factor should be adapted for each regime associated to a different slope in Figure 9. For example, it is probably possible to apply Maskell's theory (relation 5) with a blockage constant of $\varepsilon = 0.96$ for the sub-critical flow regime, when the slope is along the solid line of Figure 9, and another value of this parameter for the super-critical regime. However, up to now this value is not yet known. A more detailed investigation is under progress at the von Karman Institute to determine this empirical factor for the different flow regimes of a circular cylinder.

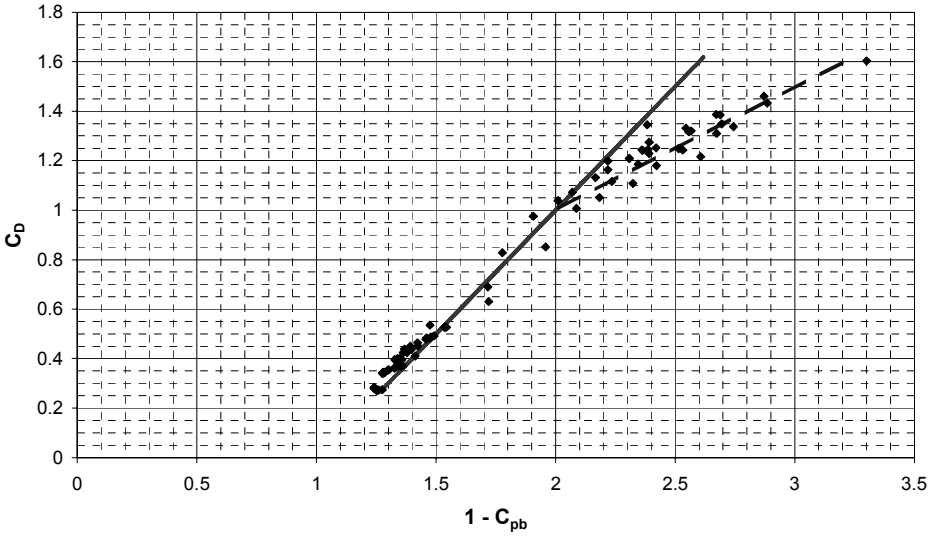


Figure 9: the drag coefficient vesus $(1 - C_{pb})$ for circular cylinders.

5 CONCLUSIONS

It is really difficult to avoid wall effects when investigating the drag coefficient of circular cylinders. Different formulations for correcting blockage effect on drag evolution are compared for each flow regime based on drag measurements of circular cylinders performed in a large low-speed wind tunnel for three different blockage ratios. None of the correction models known in the literature is valid for all the flow regimes. It is shown that the formulation providing the best correction for cylinders depends on the flow regime. To optimize the correction and reduce the scatter of the results, different correction models should be combined depending on the flow regimes. In the sub-critical regime, the best results are obtained using Allen and Vincenti's formula or Maskell's theory with $\varepsilon = 0.96$. In the super-critical regime, one should prefer using Glauert's formula with $G = 0.6$ or the model of Modi and El-Sherbiny. The change in the formulations is appearing at the flow transition with a variation of the wake pattern when passing from sub-critical to super-critical flow regimes.

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